

Nathaniel DeLeon

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Portfolio: <https://completecane98.github.io/>

EDUCATION

South Texas College

Graduation: December 2022

Associate of Science in Interdisciplinary Studies

GPA: 4.0 / 4.0

The University of Texas Rio Grande Valley

Expected Graduation: December 2026

Bachelor of Science in Computer Science

GPA: 4.0 / 4.0

SKILLS

- **Languages:** Python, C++, Dart
- **Libraries/Frameworks:** PyTorch, OpenCV, Pandas, RetinaFace, ROS, Gazebo, PyQt6, YOLO, Flutter, uFactorySDK
- **Tools:** Git/GitHub, Linux, VS Code, SolidWorks, Latex, uFactoryXarm 5

PROJECTS

UAV Natural Language Tracker

December 2025 – June 2026

- Built a Python-based UAV targeting application within gazebo simulation that utilizes natural language vocabulary and reliable video streaming (UDP Based transmission).
- Utilized **GPT 5.2** and **YOLO-World** to develop a **natural language detection pipeline**, utilizing GPT's semantic reasoning and YOLO-World's object detection
- Implemented full autonomous target tracking by mapping natural language text commands directly to real-time bounding boxes.

Remote Desktop System

October 2025 – February 2026

- Built a Python-based low-latency remote desktop application that utilizes auto-discovery and reliable video streaming (UDP Based transmission).
- Reduced latency and bandwidth using threaded **OpenCV** processing and dynamic frame-dropping.
- Implemented full remote input control (mouse/keyboard) using **pynput** and **pyautogui**.

Deepfake Detector

September 2025 – December 2025

- Engineered a hybrid spatial/temporal deep learning model combining **ResNet50** and **LSTMs** to detect video manipulation, utilizing transfer learning for robust feature extraction.
- Developed an automated video preprocessing pipeline using **RetinaFace** to detect, crop, and normalize face sequences from raw footage for consistent model input.
- Implemented a custom **PyTorch** training loop with dynamic frame sampling and validation checkpoints to optimize classification accuracy between real and fake media.
- Achieved up to 92% classification accuracy, displaying a strong ability to identify subtle spatial and temporal anomalies in synthetic media

WORK/RESEARCH EXPERIENCE

USDA REEU Program at the University of Texas at San Antonio – San Antonio, Texas

June 2026 – July 2026

- Collaborated on a **Python-based Computer Vision** pipeline for crop picking, sorting, and packaging for the University of Texas San Antonio in a partnership with the United States Department of Agriculture
- Engineered an object detection pipeline utilizing **YOLOv11** to perform real-time, multi-class crop verification. Managed the **dataset sourcing/annotation** and iteratively trained various model configurations to optimize for speed and accuracy
- Developed a custom class for a **uFactory xArm 5** to translate automated detection into mechanical movement. Mapped the physical workspace to allow for end-effector centering, collision avoidance, and continuous-path routing in grid packaging.

Teaching Assistant for Department of Computer Science – Edinburg, Texas

August 2025 – Current

- Acting Teaching Assistant for Automata and formal Languages Course at the University of Texas Rio Grande Valley. Previously taught Software Engineering 1 and Computer Science 1.
- Assisted Professor in class procedures such as management of lab sessions.
- Provide Availability to students for individual or group guidance within subject topics such as desktop-based software and web-based software and their applications.
- Grade and provide feedback for Quiz assignments.

Research Assistant for Department of Computer Science – Edinburg, Texas

May 2025 – August 2025

- Focused within Algorithms and Self Assembly, specifically studying topics such as Self Assembly and Drainability.
- Collaborated with faculty to analyze algorithmic self-assembly problems based in complexity and probabilistic bounds.

ORGANIZATIONS

Telemetry Lead - Society of Aerospace and Robotic Engineers – Edinburg, Texas

August 2025 – August 2026

- Developed project components and concepts validating the use of Visco Acoustic Electrospinning for Nanofiber Guidance.
- Delegated Team tasks to Telemetry members. These tasks contributed to the completion and success of the project.
- Managed the electronic and software components needed for project completion.
- Secured a \$15,000 prize as a finalist in the NASA ORBIT Competition by managing electronic and software project components.

AWARDS

- University of Texas Rio Grande Valley Luminary Scholar
- 6x President's List Award Recipient
- NASA ORBIT Competition Finalists